

# Postmortem: notifier reminders delayed by a stuck queue worker

|                    |                                   |
|--------------------|-----------------------------------|
| Incident date      | 2023-03-14 (Tuesday, morning JST) |
| Severity           | SEV-2                             |
| Service            | notifier                          |
| Incident commander | Aiko                              |
| Author             | Tom                               |
| Status             | Final, reviewed 2023-03-16        |

## Summary

On 2023-03-14 the morning batch of SMS reminders left up to 77 minutes late. One of the three notifier queue workers lost its queue lease every few seconds, after a config change the day before had cut the lease time from 60 s to 6 s, and the two healthy workers could not keep up with the batch. Setting the lease time back to 60 s and restarting the workers cleared the queue by 08:47.

## Impact

- Customers: 4,212 SMS reminders for appointments between 09:00 and 11:00 JST were delayed; the last one left 77 minutes late.
- 41 appointments started before their reminder arrived. Support counted 17 customer reports, most of them from businesses.
- Money: none moved. Data: none exposed or lost.
- Email confirmations were not affected; they use a separate queue.

## Timeline (JST)

| Time  | What happened                                                                                                                                                                    | Who                |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| 07:30 | The scheduler puts the morning reminder batch on the notifier queue: 4,212 SMS reminders for appointments between 09:00 and 11:00 JST.                                           | notifier scheduler |
| 07:38 | Alert fires: notifier queue depth above 2,000 for 5 minutes. The paging tool pages the on-call engineer.                                                                         | paging tool        |
| 07:41 | Page acknowledged. Ravi opens a thread in #inc-live with the alert line and a screenshot of the queue depth panel.                                                               | Ravi               |
| 07:49 | notifier logs show worker 2 of 3 repeating "lease renewal failed" about once a second and taking no new messages. Workers 1 and 3 are sending but cannot keep up with the batch. | Ravi               |
| 07:55 | First customer report through support: a hair salon says its 09:00 clients usually have their reminder by 07:45 and none has arrived.                                            | Support            |
| 08:02 | Incident declared (SEV-2). Aiko takes incident commander and pages Tom, because the queue workers are his area.                                                                  | Aiko               |

| Time  | What happened                                                                                                                                                                                        | Who  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 08:09 | Tom joins. He confirms on status.twilio.com that twilio is healthy and that messages sent by workers 1 and 3 are being delivered normally.                                                           | Tom  |
| 08:16 | Tom restarts worker 2. It takes new messages for about forty seconds, then falls back into the same lease renewal error.                                                                             | Tom  |
| 08:24 | Tom finds the cause: a config change on 2023-03-13 cut the queue lease time from 60 s to 6 s. Worker 2 runs on the node whose clock is furthest off, so it loses every lease before it can renew it. | Tom  |
| 08:31 | The lease time is set back to 60 s and all three workers are restarted. Queue depth starts to fall by about 300 messages a minute.                                                                   | Tom  |
| 08:47 | Queue depth is back under 200 and the morning batch is fully sent. The last reminder left 77 minutes late.                                                                                           | Ravi |
| 08:55 | Aiko gives support the list of 41 appointments that started before their reminder arrived, so the businesses can be told.                                                                            | Aiko |
| 09:20 | Incident closed. Aiko posts the summary in #inc-live and books the postmortem for the Thursday review.                                                                                               | Aiko |

## What went well

- The log line said "lease renewal failed" in plain words, so the stuck worker was found within eight minutes of the page.
- twilio was ruled out quickly, because two workers were still delivering.

## What went badly

- The queue depth alert fired 8 minutes after the batch started; an alert on the age of the oldest message would have said the same thing with more urgency.
- The lease time change on 2023-03-13 was reviewed as a tuning change and was not tested against a node with clock drift.

## Action items

| Action                                                                                             | Owner | Due        | Status |
|----------------------------------------------------------------------------------------------------|-------|------------|--------|
| Alert on the age of the oldest message on the notifier queue, not only on queue depth.             | Tom   | 2023-03-24 | Done   |
| Enforce a minimum lease time of 30 s in code, so a config change cannot set it lower.              | Tom   | 2023-03-31 | Done   |
| Add per-worker send rate to the "Sorrel / notifier" board, so one silent worker shows at a glance. | Ravi  | 2023-04-07 | Done   |
| Check clock sync on every notifier node in the weekly node health report.                          | Ravi  | 2023-04-14 | Open   |